

Completeness Extraction for $\mathcal{ALCI}_{\text{RCC5}}$ (v2, Repaired)

Closing the model-to-quotient gap via witness-generated submodels,
generated covers, and reachability discharge

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Abstract

This is a repaired version of the completeness-extraction note for the split-forest approach to $\mathcal{ALCI}_{\text{RCC5}}$ decidability. An earlier version [7] contained two structural defects that prevented it from establishing the model-to-quotient direction: a Hasse construction that assumed a well-founded ambient $\leq_{\text{PP}}$ DAG with a $\leq_{\text{PP}}$ -minimal root (which fails for concepts forcing infinite PP-chains, e.g. $\exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$), and a triple-coherence proof with witness conflation across distinct occurrences.

The repair follows the structural ideas of GPT 5.5’s coherent generated split-forest proof [5], but *without* introducing Büchi automata or ω -acceptance machinery. We apply seven concrete repairs:

- (R1) Pass to a *witness-generated submodel* (sparse, logic-generated, no density issue);
- (R2) Use *generated cover edges* (selected witness-cover edges in the generated submodel), not semantic Hasse covers;
- (R3) Drop the ambient-root requirement; permit *self-loops on the parent function* of the finite quotient, so ultimately periodic ancestor trunks are first-class;
- (R4) Make the *containment orientation* explicit (parent = proper superpart, child = proper part);
- (R5) Replace raw pair-tables with *support-closed mosaics* that store joint atomic networks, fixing the C5 witness conflation;
- (R6) Add an explicit *typed-EQ congruence* axiom, separate from blocking equivalence;
- (R7) Discharge transitive PP/PPI eventualities by *finite-graph reachability* on the finite quotient.

The crucial observation is that the cycle-realization condition of [5] reduces to ordinary reachability over the finite quotient: $\exists \text{PP}.D \in \lambda(\sigma)$ requires $D \in \lambda(\sigma')$ for some σ' reachable from σ along parent edges (including self-loops). This is decidable in $O(|S|^2)$ time per eventuality and requires no ω -language machinery.

Combined with GPT-5.4’s soundness chain [4], this yields decidability of concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ under strong EQ semantics.

1 Introduction

The split-forest proposal for $\mathcal{ALCT}_{\text{RCC5}}$ decidability [4] proves the soundness direction in full detail: every valid finite rank- d quotient unfolds into a genuine strong-EQ model via canonical refinements of disjunctive RCC5 networks. The completeness direction—that every satisfiable concept admits a valid finite quotient—was stated as a proof sketch.

A first attempt to write out this completeness extraction (“v1”) [7] appeared in April 2026. Subsequent review by GPT-5.5 identified two structural defects:

Defect 1: Hasse construction with rooted-tree assumption. v1 began by orienting the proper-part relation $\leq_{\text{PP}}$ as a DAG and choosing a $\leq_{\text{PP}}$ -minimal element as ambient root. This works when the model has a unique minimum or when one can be added freshly. But the concept

$$C \uparrow := \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

forces a strictly increasing PP-chain $d_0 \text{PP} d_1 \text{PP} d_2 \text{PP} \dots$ with no $\leq_{\text{PP}}$ -minimum. Adding a fresh top via $\rho(d_\top, d) = \text{PP}$ *inverts* the orientation: by RCC5 semantics, $d_\top \text{PP} d$ means $d_\top$ is a proper part of d , not the other way around. v1’s construction therefore breaks on any concept forcing an infinite PP-chain.

Defect 2: Witness conflation in (C5) triple coherence. v1’s proof of (C5) implicitly combined witnesses from different occurrences as if they formed a coherent triple. Raw pair tables admit independent witnesses for $M(\chi_1, \chi_2)$ and $M(\chi_1, \chi_3)$ that may not be jointly realizable. Triple coherence requires a single triple (x_1, x_2, x_3) realizing all three relations simultaneously; raw tables do not enforce this.

GPT-5.5’s coherent generated split-forest proof [5] repairs both defects. We follow its approach with one important difference: where GPT-5.5 phrases the eventuality discharge in automata-theoretic language (“cycle-realization condition”, Definition 6.4 of that paper), we observe that this condition reduces to *finite-graph reachability over the finite quotient*, and we present the proof in that finitary style.

1.1 Plan

Section 2 recaps types, closure, need families, and the model-edge relation-safety lemma. Section 3 introduces the witness-generated submodel reduction. Section 4 fixes the containment orientation and defines generated cover edges. Section 5 constructs the generated split-tree presentation, with no ambient root and self-loops permitted on the parent function. Section 6 defines rank- d profiles and the finite quotient. Section 7 extracts support-closed sibling-interface mosaics. Section 8 defines witness menus for DR/PO existentials. Section 9 states the typed-EQ congruence axiom. Section 10 verifies (C1)–(C5). Section 11 proves reachability discharge for PP/PPI eventualities. Section 12 assembles validity. Section 13 states the main theorem and decidability. Section 14 compares with GPT-5.5’s automata version and discusses honest scope.

1.2 Main result

Theorem 1.1 (Completeness extraction). *If $\mathcal{I}, d_* \models C_0$ in $\mathcal{ALCI}_{\text{RCC5}}$ under strong EQ semantics, then there exists a valid finite generated split-forest quotient Q for C_0 with $d = \text{md}(C_0)$.*

Combined with the soundness chain of [4, 5], this yields:

Corollary 1.2 (Decidability of $\mathcal{ALCI}_{\text{RCC5}}$). *Concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ under strong EQ semantics is decidable.*

2 Preliminaries

We work under the conventions of [4]. This section recalls only the notions used in the extraction.

2.1 Types and closure

Fix a concept C_0 in negation normal form (NNF). The Fischer-Ladner closure $\text{cl}(C_0)$ is the smallest set containing C_0 closed under subformulas and NNF complements. A *Hintikka type* $\tau \in \text{Tp}(C_0) \subseteq 2^{\text{cl}(C_0)}$ is a maximally consistent subset satisfying the standard propositional closure conditions. For a in a model $\mathcal{I}$, write $\text{tp}_{\mathcal{I}}(a) = \{D \in \text{cl}(C_0) : \mathcal{I}, a \models D\}$; this is a Hintikka type.

2.2 Need families and relation-safety

Definition 2.1 (Per-relation needs). For $\tau \in \text{Tp}(C_0)$ and $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$, define

$$\text{Need}_R(\tau) := \{D \in \text{cl}(C_0) : \forall R.D \in \tau\}.$$

A typed edge (τ, R, σ) with $\tau, \sigma \in \text{Tp}(C_0)$ is *relation-safe* when

$$\text{Need}_R(\tau) \subseteq \sigma \quad \text{and} \quad \text{Need}_{\text{inv}(R)}(\sigma) \subseteq \tau.$$

Lemma 2.2 (Model edges are relation-safe). *In any model $\mathcal{I}$ of $\mathcal{ALCI}_{\text{RCC5}}$, if $\rho(a, b) = R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ for distinct elements a, b , then $(\text{tp}_{\mathcal{I}}(a), R, \text{tp}_{\mathcal{I}}(b))$ is relation-safe.*

Proof. If $\forall R.D \in \text{tp}_{\mathcal{I}}(a)$, then $\mathcal{I}, a \models \forall R.D$ and $\rho(a, b) = R$, hence $\mathcal{I}, b \models D$, so $D \in \text{tp}_{\mathcal{I}}(b)$. The dual inclusion follows by applying the same argument with the inverse relation at b . $\square$

This elementary observation is the engine of the entire extraction: every relation realized in the model is automatically relation-safe for its endpoints' types.

2.3 RCC5 composition

We use the standard RCC5 composition table $\text{comp}(R, S) \subseteq \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$ (see [5, Table 1]). The composition is sound for any abstract RCC5 frame: $\rho(a, c) \in \text{comp}(\rho(a, b), \rho(b, c))$.

3 Witness-generated submodel reduction

The first repair is to restrict attention to sparse, logic-generated submodels. This sidesteps every concern about arbitrary-density semantic orders and lets us define cover edges relative to a finite generated set.

Definition 3.1 (Selected witness policy). Let $\mathcal{I}$ be a model. A *selected witness policy* w chooses, for every $a \in \Delta^{\mathcal{I}}$ and every existential $\exists R.D_0 \in \text{cl}(C_0)$ with $a \in (\exists R.D_0)^{\mathcal{I}}$, one element

$$w(a, \exists R.D_0) \in \Delta^{\mathcal{I}} \quad \text{with} \quad \rho(a, w(a, \exists R.D_0)) = R \text{ and } w(a, \exists R.D_0) \in D_0^{\mathcal{I}}.$$

We write $a \xrightarrow{R}_w b$ when b is so selected.

Definition 3.2 (Witness-generated submodel). Let $a_0 \in \Delta^{\mathcal{I}}$. The *witness-generated domain* $D_w(a_0) \subseteq \Delta^{\mathcal{I}}$ is the smallest set containing a_0 and closed under all selected witnesses for closure-existentials true at its members. The *induced submodel* $\mathcal{I}_w$ has domain $D_w(a_0)$ with $\rho^{\mathcal{I}_w}$ and atomic concept interpretations restricted from $\mathcal{I}$.

Lemma 3.3 (Witness-generated submodel lemma). *If $a_0 \in C_0^{\mathcal{I}}$, then for every $a \in D_w(a_0)$ and every $E \in \text{cl}(C_0)$,*

$$a \in E^{\mathcal{I}} \iff a \in E^{\mathcal{I}_w}.$$

In particular, $a_0 \in C_0^{\mathcal{I}_w}$.

Proof. The induced substructure remains a strong abstract RCC5 frame: all RCC5 frame requirements (irreflexivity of $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$, converse, composition) are universal over retained pairs and triples and are preserved under restriction. Atomic concepts are preserved by restriction.

Boolean cases of the equivalence are immediate from the inductive hypothesis.

For $E = \exists R.F$: if $a \in E^{\mathcal{I}_w}$, retention into $\mathcal{I}_w$ implies the witness exists in $\mathcal{I}$, so $a \in E^{\mathcal{I}}$. Conversely, if $a \in E^{\mathcal{I}}$, the selected witness $b = w(a, \exists R.F)$ lies in $D_w(a_0)$ and satisfies F in $\mathcal{I}$; by induction $b \in F^{\mathcal{I}_w}$, so $a \in E^{\mathcal{I}_w}$.

For $E = \forall R.F$: if $a \in E^{\mathcal{I}}$, every retained R -neighbour of a is also an R -neighbour in $\mathcal{I}$ and satisfies F ; by induction it satisfies F in $\mathcal{I}_w$. Conversely, suppose $a \notin E^{\mathcal{I}}$. Then $a \in (\exists R.\bar{F})^{\mathcal{I}}$, where $\bar{F}$ is the NNF complement, and $\exists R.\bar{F} \in \text{cl}(C_0)$ because the closure is closed under NNF complements. The selected witness for $\exists R.\bar{F}$ is retained and, by induction, satisfies $\bar{F}$ in $\mathcal{I}_w$; hence $a \notin E^{\mathcal{I}_w}$. $\square$

Corollary 3.4 (Sparse generated models suffice). *Every satisfiable C_0 has a witness-generated model with $|D_w(a_0)| \leq \aleph_0$. The cover-tree extraction below need only address this sparse class.*

Remark 3.5 (Why the density issue dissolves). Lemma 3.3 replaces v1's flawed Hasse construction with a model-theoretic thinning. The resulting $\mathcal{I}_w$ contains exactly the elements that the logic witnesses for: the evaluation point and selected witnesses for existentials in the closure. Density of an arbitrary semantic PP/PPI order is irrelevant, because we never quantify over it.

4 Generated cover edges and containment orientation

We now fix the orientation convention used throughout the rest of the paper.

Containment orientation. *Parent* = proper superpart; *child* = proper part.
Equivalently: along a tree branch, child PP parent and parent PPI child.

This convention determines which existentials produce “upward” moves and which produce “downward” moves in the split tree. $\exists \text{PP}.D$ at a requires a proper-superpart witness, hence an *ancestor* in the tree. $\exists \text{PPI}.D$ at a requires a proper-part witness, hence a *descendant*.

Definition 4.1 (Generated cover edge). Let $A \subseteq D_w(a_0)$ be a finite witness-generated subset of the submodel. A *generated PPI-cover edge* $x \xrightarrow{\text{PPI}}_{\text{gen},A} y$ holds when:

- $x, y \in A$ and $\rho(x, y) = \text{PPI}$ (i.e., y is a proper part of x);
- no element $z \in A$ satisfies $\rho(x, z) = \text{PPI}$ and $\rho(z, y) = \text{PPI}$ (no selected element of A lies strictly between x and y in the proper-part order).

The dual generated PP-cover edge $x \xrightarrow{\text{PP}}_{\text{gen},A} y$ is defined symmetrically.

Remark 4.2 (Generated covers are not Hasse covers). A generated cover edge is immediate relative to the finite selected set A . It is not claimed to be an immediate cover in the full ambient PP-order. Repaired wording from the survival note [6]:

A cover edge is a generated witness-cover edge, not a semantic Hasse cover.

Constructing covers relative to A is well-defined even when the ambient order is dense or has no Hasse covers at all; we simply pick the immediate-in- A relation. Crucially, A grows with the generation depth, but at every finite depth the relative cover relation is well-defined.

5 Generated split-tree presentation

We construct a (possibly infinite) split tree from a witness-generated model. No ambient root is required.

Construction 5.1 (Generated split-tree). Let $\mathcal{I}_w$ be a witness-generated model with evaluation point a_0 . Build a tree T of *occurrences* together with maps

$$\text{orig} : T \rightarrow D_w(a_0), \quad \lambda : T \rightarrow \text{Tp}(C_0), \quad \text{par} : T \rightarrow T,$$

inductively as follows.

1. **Seed.** Create a root occurrence u_0 with $\text{orig}(u_0) = a_0$ and $\lambda(u_0) = \text{tp}_{\mathcal{I}_w}(a_0)$.
2. **Upward expansion (PP-witnesses).** For each occurrence u and each $\exists \text{PP}.D \in \lambda(u)$:
 - If $\text{par}(u)$ is undefined or $\lambda(\text{par}(u)) \not\preceq D$, select a witness $b = w(\text{orig}(u), \exists \text{PP}.D) \in D_w(a_0)$ with $\rho(\text{orig}(u), b) = \text{PP}$ and create a new occurrence $u^\uparrow$ with $\text{orig}(u^\uparrow) = b$, $\lambda(u^\uparrow) = \text{tp}_{\mathcal{I}_w}(b)$, $\text{par}(u) = u^\uparrow$.

- If a parent already records D , no new occurrence is needed.
3. **Downward expansion (PPI-witnesses).** For each occurrence u and each $\exists \text{PPI}.D \in \lambda(u)$, select a witness $b = w(\text{orig}(u), \exists \text{PPI}.D)$ and create a child occurrence $u^\downarrow$ with $\text{orig}(u^\downarrow) = b$, $\lambda(u^\downarrow) = \text{tp}_{\mathcal{I}_w}(b)$, and $\text{par}(u^\downarrow) = u$.
 4. **Sibling expansion (DR/PO-witnesses).** For each occurrence u and each $\exists R.D \in \lambda(u)$ with $R \in \{\text{DR}, \text{PO}\}$, select a witness $b = w(\text{orig}(u), \exists R.D)$ and record an outgoing R -witness edge from u to a node mapped to b (typically a sibling of u under a common ancestor).
 5. **Splitting.** If a single element $b \in D_w(a_0)$ must appear under multiple distinct parents in this construction, introduce one occurrence per containment branch. These occurrences are declared EQ-mates (typed equivalence classes under the relation defined in Section 9).
 6. **Edge labels.** For comparable occurrences u, v on the same branch with $v = \text{par}^k(u)$ for some $k > 0$, set $\rho_T(u, v) = \text{PP}$ and $\rho_T(v, u) = \text{PPI}$ (transitively closed along the branch). For incomparable occurrences, $\rho_T(u, v)$ is determined by the realized RCC5 relation $\rho(\text{orig}(u), \text{orig}(v))$ in $\mathcal{I}_w$.

Remark 5.2 (Branches need not terminate). The parent chain of an occurrence may be:

- finite, terminating at an occurrence with no $\exists \text{PP}.D$ in its type;
- one-sided infinite (e.g., for $C^\uparrow$);
- bi-infinite (e.g., for concepts forcing both unbounded PP and unbounded PPI chains).

We do *not* require a $\leq_{\text{PP}}$ -minimal element or any ambient top. In the finite quotient (Section 6), infinite parent chains are represented by self-loops or longer cycles on the parent function.

Lemma 5.3 (Existence). *Every witness-generated model $(\mathcal{I}_w, a_0)$ admits a generated split tree as in Construction 5.1.*

Proof. Construct occurrences coinductively. At every step, the existing occurrences are finite (since each step adds finitely many witnesses for the closure-existentials of one type). The construction is well-defined because $\text{cl}(C_0)$ is finite, and there are only finitely many distinct closure-existentials at any type. Pull-back of ρ via orig inherits converse and composition. Type assignments inherit Hintikka conditions from $\mathcal{I}_w$ via Lemma 3.3. $\square$

6 Rank- d profiles and the finite quotient

We now define the finite combinatorial object that summarizes the (possibly infinite) split tree.

Definition 6.1 (Profile components). For an occurrence $u \in T$ at rank k , the *rank- k profile* $\text{st}_k(u)$ records:

1. the closure type $\lambda(u)$;
2. the *typed-EQ colour* $c_{\text{EQ}}(u)$ (a label from a finite alphabet, used to track EQ-mate classes);
3. the rank- $(k-1)$ profile of $\text{par}(u)$ (with $\perp$ if undefined);

4. the rank- $(k-1)$ *child multiplicities*: a map $\Sigma_{k-1} \rightarrow \{0, 1, 2+\}$ counting children by their rank- $(k-1)$ profile;
5. the rank- $(k-1)$ *incident sibling-interface mosaics* for ordered pairs of children (Section 7);
6. the rank- $(k-1)$ *witness menu* for DR/PO existentials (Section 8);
7. the *vertical eventuality summary*: for each $\exists\text{PP}.D \in \lambda(u)$, the set of ranks at which D appears in the parent chain; symmetric for $\exists\text{PPI}.D$.

A rank-0 profile is just a Hintikka type. The finite profile alphabet Σ_d at rank $d = \text{md}(C_0)$ has bounded cardinality (Lemma 6.2).

Lemma 6.2 (Finite-index). *For each $k \leq d$, the set Σ_k of rank- k profiles is finite, and $|\Sigma_k|$ is bounded by an effectively computable function of $|\text{cl}(C_0)|$.*

Proof. Identical to the finite-index argument in [5, Lemma 7.4]: closure types are finite; typed-EQ colours are drawn from a finite palette of size $O(2^{|\text{cl}(C_0)|})$; child multiplicities are truncated at $2+$ (threshold counting); incident mosaics are bounded sets of bounded networks over a finite endpoint-summary alphabet (Section 7); witness-menu entries are pairs from a finite set; eventuality summaries are subsets of $\text{cl}(C_0)$. Induction on k closes the bound. $\square$

Definition 6.3 (Finite quotient). The *finite generated split-forest quotient* of T is

$$Q = (S, \lambda_Q, \text{par}_Q, \text{Child}_Q, \text{Iface}_Q, \text{Wit}_Q, c_{\text{EQ}, Q}, \sigma_*)$$

where:

- $S = \{\text{st}_d(u) : u \in T\} \subseteq \Sigma_d$;
- $\lambda_Q(\sigma)$ is the closure type at σ ;
- $\text{par}_Q : S \rightarrow S$ is the projected parent function: $\text{par}_Q(\sigma) = \text{st}_d(\text{par}(u))$ for any $u \in T$ with $\text{st}_d(u) = \sigma$ (well-defined by profile equality, including the parent slot; $\text{par}_Q(\sigma) = \sigma$ is permitted, as is $\text{par}_Q(\sigma) = \perp$);
- $\text{Child}_Q, \text{Iface}_Q, \text{Wit}_Q$ are inherited from the profile components;
- $c_{\text{EQ}, Q}$ is the projected typed-EQ colouring;
- $\sigma_* = \text{st}_d(u_*)$ for the occurrence u_* with $\text{orig}(u_*) = a_0$.

Remark 6.4 (Self-loops are allowed). The parent function par_Q may have self-loops: $\text{par}_Q(\sigma) = \sigma$. This represents an infinite upward PP-chain that, at rank d , has stabilized to a single profile. The unfolding interprets each traversal of the self-loop as a fresh occurrence. Concretely, for $C \uparrow= \exists\text{PP}.\top \sqcap \forall\text{PP}.\exists\text{PP}.\top$, the quotient has a single profile σ with $\text{par}_Q(\sigma) = \sigma$ and $\lambda_Q(\sigma) \ni \exists\text{PP}.\top$, $\lambda_Q(\sigma) \ni \forall\text{PP}.\exists\text{PP}.\top$.

Crucially, $\text{par}_Q(\sigma) = \sigma$ **does not mean** “ σ is its own parent in the model.” It means: the parent occurrence has the same rank- d profile. In the unfolding, every traversal produces a distinct semantic occurrence. This is a blocking cycle, not an EQ-identification.

Lemma 6.5 (Finiteness of Q). $|S| \leq |\Sigma_d| < \infty$.

7 Sibling-interface descriptors as support-closed mosaics

This section replaces v1's raw pair-tables with support-closed mosaics, fixing the C5 witness conflation.

7.1 Status partition

For an occurrence u with two distinct child subtrees T_s, T_t at children s, t , we partition descendants of s by their RCC5 relation to t :

Definition 7.1 (Status partition). For x a descendant of s (including s itself):

- $(x) = \text{Core}$ if $\text{orig}(x)$ lies below both $\text{orig}(s)$ and $\text{orig}(t)$ in the generated PPI-order, or is an EQ-mate visible to both sides;
- $(x) = \text{Out}$ if $x \notin \text{Core}$ and $\rho(\text{orig}(x), \text{orig}(t)) = \text{DR}$;
- $(x) = \text{Front}$ if $x \notin \text{Core}$ and $\rho(\text{orig}(x), \text{orig}(t)) = \text{PO}$.

Lemma 7.2 (Three-way partition). *Every descendant x of s has exactly one status in $\{\text{Core}, \text{Out}, \text{Front}\}$. Symmetrically for descendants of t .*

Proof. By Lemma 7.3 and the open cross-edge restriction of [4, Theorem 1.4]: if x is a non-core descendant of s , then $\rho(\text{orig}(x), \text{orig}(t)) \in \{\text{DR}, \text{PO}\}$ (it cannot be PP, PPI, or EQ without forcing x into Core via generated containment closure). $\square$

Lemma 7.3 (Open cross-edge). *If x is a non-core descendant of s and y is a non-core descendant of t , then $\rho(\text{orig}(x), \text{orig}(y)) \in \{\text{DR}, \text{PO}\}$.*

Proof. EQ would force both into the typed-EQ congruence and hence into Core. PP would force x into the generated PP-closure of material below $\text{orig}(t)$, hence visible to t , contradicting non-coreness. PPI is symmetric. $\square$

7.2 Endpoint summaries and atomic mosaics

Definition 7.4 (Endpoint summary). An *endpoint summary* at rank k consists of:

1. a rank- k profile $p \in \Sigma_k$;
2. a status $\sigma \in \{\text{Core}, \text{Out}, \text{Front}\}$;
3. a typed-EQ colour $c_{\text{EQ}} \in \text{Colours}$.

The set E_k of endpoint summaries is finite.

Definition 7.5 (Atomic support mosaic). An *atomic support mosaic* on a finite set V of vertices, each labelled by an endpoint summary, is a complete labelling $\mu : V \times V \rightarrow \{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$ such that:

- (M1) $\mu(v, v) = \text{EQ}$ for all v ;
- (M2) $\mu(w, v) = \text{inv}(\mu(v, w))$;
- (M3) $\mu(u, w) \in \text{comp}(\mu(u, v), \mu(v, w))$ (RCC5 composition);

- (M4) every edge is relation-safe for the closure types of its endpoint summaries (Definition 2.1);
- (M5) **Out**-status endpoints have only **DR** as cross-status outgoing label;
- (M6) non-core **Front–Front** cross-status edges have labels in $\{\mathbf{DR}, \mathbf{PO}\}$;
- (M7) endpoints with the same typed-EQ colour have label **EQ** (typed-EQ congruence respects the mosaic).

Definition 7.6 (Sibling descriptor). A *rank- k sibling descriptor* for an ordered child pair (s, t) is a finite set $\mathbf{Mos}_k(s, t)$ of atomic support mosaics over endpoint summaries at rank k , capturing the joint atomic networks that occur in the model among finitely many endpoints in $T_s \cup T_t \cup \{s, t\}$.

The descriptor is *support-closed* when it is closed under:

- (SC1) **Restriction.** Every induced submosaic on a vertex subset is in $\mathbf{Mos}_k(s, t)$.
- (SC2) **Triple composition.** If a 3-vertex submosaic on (v_1, v_2, v_3) has labels (R_{12}, R_{13}, R_{23}) , then $R_{13} \in \text{comp}(R_{12}, R_{23})$.
- (SC3) **Typed-EQ replacement.** Replacing a vertex by an EQ-mate of the same endpoint summary yields a mosaic in $\mathbf{Mos}_k(s, t)$.
- (SC4) **One-step containment extension.** Adding a generated PPI-child of a vertex (with appropriate label propagation by composition with **PP** or **PPI**) yields a mosaic in $\mathbf{Mos}_k(s, t)$.

Lemma 7.7 (Support closure is finite). *For fixed rank k , the support closure of any finite set of mosaics is finite and effectively computable.*

Proof. The endpoint summary alphabet E_k is finite. Mosaics are bounded in size by some constant depending only on $|\text{cl}(C_0)|$ (we may restrict to mosaics on at most $|\text{cl}(C_0)| + 4$ vertices, since larger mosaics can be decomposed into overlapping smaller ones via restriction). Each closure operation adds elements from this finite universe; iteration terminates. $\square$

7.3 Extraction from a model

Construction 7.8 (Mosaic extraction). For each ordered child pair (s, t) in the split tree at rank k :

1. For every finite vertex set $V \subseteq T_s \cup T_t \cup \{s, t\}$, form the *realized mosaic* μ_V with $\mu_V(v, w) = \rho(\text{orig}(v), \text{orig}(w))$ and endpoint summaries given by the rank- k profile, status, and typed-EQ colour of each $v \in V$.
2. Initialize $\mathbf{Mos}_k^{\text{raw}}(s, t)$ as the set of realized mosaics for V ranging over finite subsets up to a bounded vertex count.
3. Close under (SC1)–(SC4) of Definition 7.6.

The result $\mathbf{Mos}_k(s, t)$ is the *extracted sibling descriptor*.

Lemma 7.9 (Extracted descriptors are coherent). *The descriptor $\mathbf{Mos}_k(s, t)$ extracted in Construction 7.8 satisfies (M1)–(M7) and (SC1)–(SC4).*

Proof. (M1)–(M3) hold for realized mosaics because $\mathcal{I}_w$ is a strong abstract RCC5 frame (Section 2); they are preserved under (SC1)–(SC4) closure operations. (M4) holds by Lemma 2.2. (M5)–(M6) hold by Lemma 7.2 and Lemma 7.3. (M7) holds by the definition of the typed-EQ colouring (Section 9). (SC1)–(SC4) hold by construction (we explicitly close). $\square$

8 Witness menus for DR/PO existentials

For each rank- d profile $\sigma \in S$ and each $\exists R.D \in \lambda_Q(\sigma)$ with $R \in \{\text{DR}, \text{PO}\}$, the witness menu records a target profile and relation.

Construction 8.1 (Witness-menu extraction). For each $u \in T$ with $\text{st}_d(u) = \sigma$ and each $\exists R.D \in \lambda(u)$ with $R \in \{\text{DR}, \text{PO}\}$:

1. Locate the selected witness $b = w(\text{orig}(u), \exists R.D) \in D_w(a_0)$ from the witness policy.
2. Find a tree occurrence v with $\text{orig}(v) = b$ (such v exists by Construction 5.1, possibly as a sibling occurrence under a common ancestor).
3. Record the witness-menu entry $(\exists R.D) \mapsto (\text{st}_d(v), R)$ in $\text{Wit}_d(\sigma)$.

Lemma 8.2 (Witness-menu relation-safety). *Every entry $(\exists R.D) \mapsto (\sigma', R)$ in $\text{Wit}_d(\sigma)$ satisfies: $\text{Need}_R(\lambda_Q(\sigma)) \subseteq \lambda_Q(\sigma')$ and $\text{Need}_{\text{inv}(R)}(\lambda_Q(\sigma')) \subseteq \lambda_Q(\sigma)$.*

Proof. By Lemma 2.2 applied to the model edge $(\text{orig}(u), \text{orig}(v))$ with relation R , and the fact that $\lambda_Q(\sigma)$ depends only on σ (the closure type is the first profile component). $\square$

Remark 8.3. PP and PPI existentials do *not* appear in the witness menu. They are discharged by reachability over the parent function of Q (Section 11). This separation is deliberate: PP/PPI witnesses are tracked structurally by the tree backbone, while DR/PO witnesses are tracked combinatorially by the menu.

9 Typed-EQ congruence

The typed-EQ congruence is an explicit axiom on the quotient, *separate* from blocking equivalence and from the parent function’s self-loops.

Definition 9.1 (Typed-EQ congruence). An equivalence relation $\equiv_{\text{EQ}}$ on occurrences of T (or on profiles in S , projecting from T) is a *typed-EQ congruence* when, for all x, y with $x \equiv_{\text{EQ}} y$:

- (EQ1) $\lambda(x) = \lambda(y)$ (same closure type);
- (EQ2) $c_{\text{EQ}}(x) = c_{\text{EQ}}(y)$ (same typed-EQ colour);
- (EQ3) the vertical eventuality summaries (rank- k profile components 7) at x and y agree;
- (EQ4) for every other equivalence class $[z]$, the relation $\rho(\text{orig}(x), \text{orig}(z'))$ is the same for all $z' \in [z]$ and matches $\rho(\text{orig}(y), \text{orig}(z'))$.

Remark 9.2 (Blocking versus equality). A profile loop $\text{par}_Q(\sigma) = \sigma$ (Remark 6.4) is *not* a typed-EQ congruence on the unfolding: distinct unfolding occurrences of σ have the same profile but

are *not* EQ-related. The unfolding of $C \uparrow$ has $u_0 \text{PP} u_1 \text{PP} u_2 \text{PP} \dots$ with all u_i distinct semantic occurrences sharing one profile; their typed-EQ colours form a single class only because the typed-EQ colouring assigns equal colours iff the elements are genuine EQ-mates in $\mathcal{I}_w$, not iff they have equal profiles.

Lemma 9.3 (Extracted congruence). *The relation on T defined by $u \equiv_{\text{EQ}} v$ iff $\text{orig}(u) = \text{orig}(v)$ is a typed-EQ congruence in the sense of Definition 9.1.*

Proof. (EQ1): same origin implies same type. (EQ2): the typed-EQ colour is computed from the origin (a finite refinement of the type, choosing one colour per EQ-class in $\mathcal{I}_w$). (EQ3): vertical summaries are computed from the origin's ancestor/descendant relations in $\mathcal{I}_w$, which are identical for EQ-related origins. (EQ4): if $\text{orig}(u) = \text{orig}(v)$, then for any z' , $\rho(\text{orig}(u), \text{orig}(z')) = \rho(\text{orig}(v), \text{orig}(z'))$ trivially. $\square$

10 Local coherence (C1)–(C5)

We verify the (C1)–(C5) coherence conditions of [4, Definition 1.8], now restated for mosaics.

Theorem 10.1 (Local coherence). *The descriptors $\text{Mos}_k(s, t)$ extracted in Construction 7.8 satisfy local coherence (C1)–(C5).*

Proof. **(C1) Out rigidity.** By (M5) of Definition 7.5, every Out-status endpoint contributes only DR-labelled cross-status edges.

(C2) Front–Front type-safety. Let $\mu \in \text{Mos}_k(s, t)$ contain endpoints v_1, v_2 with (Front, Front) status. By (M4), $\mu(v_1, v_2) \in \{\text{DR}, \text{PO}\}$ is relation-safe for the closure types $(\lambda(v_1), \lambda(v_2))$.

(C3) Left downward support. Suppose $\mu \in \text{Mos}_k(s, t)$ contains v_1 on the s -side, and let v'_1 be a generated PPI-child of $\text{orig}(v_1)$ (equivalently, a child occurrence in T_s). By (SC4) (one-step containment extension), the descriptor $\text{Mos}_k(s, t)$ contains a mosaic μ' extending μ with v'_1 added. The label $\mu'(v'_1, v_2)$ lies in $\text{comp}(\text{PPI}, \mu(v_1, v_2))$, hence the required composition rule holds.

(C4) Right downward support. Symmetric to (C3) on the t -side.

(C5) Triple coherence. This is where v1's witness conflation occurred. Let $v_1, v_2, v_3 \in V(\mu)$ be three vertices in a mosaic $\mu \in \text{Mos}_k(s, t)$ with the relevant statuses. Because μ is an *atomic mosaic* (a joint network on V , not a collection of independent pair labels), it directly satisfies $\mu(v_1, v_3) \in \text{comp}(\mu(v_1, v_2), \mu(v_2, v_3))$ by (M3).

The crucial point: the three relations $\mu(v_1, v_2), \mu(v_2, v_3), \mu(v_1, v_3)$ are realized *by a single triple of witnesses* ($\text{orig}(v_1), \text{orig}(v_2), \text{orig}(v_3)$) in $\mathcal{I}_w$, not by three independent pairs. This is enforced by Construction 7.8: a realized mosaic comes from a single vertex set V , with all pairs labelled by the *same* model.

Support closure (SC1)–(SC4) propagates this joint realizability: restriction preserves it (a sub-triple of a real triple is a real triple); typed-EQ replacement preserves it (an EQ-mate is realized by the same model element); one-step extension preserves it (composition through the model). $\square$

Remark 10.2 (How v1’s bug is repaired). v1 wrote “ $M((\chi_1, \chi_2))$ contains R_{12} from witnesses (x_1, x_2) , $M((\chi_1, \chi_3))$ contains R_{13} from witnesses (x'_1, x_3) , hence $\rho(\text{orig}(x_2), \text{orig}(x_3)) \in \text{comp}(\text{inv}(R_{12}), R_{13})$ appears in $M((\chi_2, \chi_3))$.” The bug: x_1 and x'_1 may be distinct occurrences with the same profile but different concrete witnesses; the composition does *not* apply across them.

The repair: mosaics record *joint* networks on a single vertex set. Triple coherence is automatic because the three relations are realized by one triple of model elements, by construction.

11 Reachability discharge for PP/PPI eventualities

This section provides the finitary alternative to GPT-5.5’s cycle-realization condition [5, Definition 6.4].

Definition 11.1 (Parent reachability in Q). For $\sigma, \sigma' \in S$, write $\sigma \rightsquigarrow_{\text{PP}} \sigma'$ if there exist $\sigma_0, \sigma_1, \dots, \sigma_n$ in S with $\sigma_0 = \sigma$, $\sigma_n = \sigma'$, $n \geq 1$, and $\text{par}_Q(\sigma_i) = \sigma_{i+1}$ for $0 \leq i < n$. Self-loops are permitted: $\text{par}_Q(\sigma) = \sigma$ implies $\sigma \rightsquigarrow_{\text{PP}} \sigma$.

Symmetrically, $\sigma \rightsquigarrow_{\text{PPI}} \sigma'$ via the inverse parent function (i.e., child function), allowing self-loops.

Lemma 11.2 (Reachability discharge). *Let Q be the quotient extracted in Construction 5.1 from a witness-generated model $(\mathcal{I}_w, a_0)$. For every $\sigma \in S$ and every $\exists R.D \in \lambda_Q(\sigma)$ with $R \in \{\text{PP}, \text{PPI}\}$, there exists $\sigma' \in S$ with $\sigma \rightsquigarrow_R \sigma'$ and $D \in \lambda_Q(\sigma')$.*

Proof. We treat $R = \text{PP}$; $R = \text{PPI}$ is symmetric.

Let $\sigma \in S$ with $\exists \text{PP}.D \in \lambda_Q(\sigma)$. Pick any occurrence $u \in T$ with $\text{st}_d(u) = \sigma$. By Construction 5.1 step 2 (upward expansion), the parent chain of u contains an occurrence u' with $D \in \lambda(u')$ (the witness for $\exists \text{PP}.D$).

Concretely: there is a finite parent chain $u = u_0, u_1, \dots, u_m$ in T with $\text{par}(u_i) = u_{i+1}$ and $D \in \lambda(u_m)$. Projecting to profiles via st_d :

$$\sigma = \text{st}_d(u_0) \xrightarrow{\text{par}_Q} \text{st}_d(u_1) \xrightarrow{\text{par}_Q} \dots \xrightarrow{\text{par}_Q} \text{st}_d(u_m).$$

This gives $\sigma \rightsquigarrow_{\text{PP}} \sigma'$ with $\sigma' = \text{st}_d(u_m)$ and $D \in \lambda_Q(\sigma')$. □

Remark 11.3 (This is finite-graph reachability, not Büchi). Lemma 11.2 reduces the discharge of transitive PP/PPI eventualities to ordinary reachability over the finite directed graph (S, par_Q) , decidable in $O(|S| + |E_Q|)$ time per eventuality (e.g., breadth-first search).

GPT-5.5’s Definition 6.4 phrases the condition in terms of stem- and-cycle decompositions of branches, with realization required on the cycle. When the branch is summarized by a finite quotient Q , “cycle in the unfolding” becomes “cycle in par_Q ”. Realization on the cycle then becomes existence of a profile in the strongly connected component containing σ that has D in its type.

Restated as graph reachability: $\exists \text{PP}.D \in \lambda_Q(\sigma)$ holds iff D appears in some profile reachable from σ via par_Q . This is decidable in polynomial time without Büchi acceptance, ω -words, or any automata-theoretic machinery.

Lemma 11.4 (Universal safety propagation). *For every $\sigma \in S$ and every $\forall R.D \in \lambda_Q(\sigma)$ with $R \in \{\text{PP}, \text{PPI}\}$, every σ' with $\sigma \rightsquigarrow_R \sigma'$ has $D \in \lambda_Q(\sigma')$.*

Proof. By Lemma 2.2 applied along the parent chain in $\mathcal{I}_w$: every PP-related pair satisfies the universal, hence its target type contains D . Profile equality preserves type, so $\lambda_Q(\sigma') \ni D$. $\square$

12 Validity of the extracted quotient

We verify that the quotient extracted in Definition 6.3 is valid in the sense of [4, Definition 1.10] (or equivalently of [5, Definition 8.2]).

Theorem 12.1 (Validity). *The quotient Q extracted from $(\mathcal{I}_w, a_0)$ in Definition 6.3 satisfies all validity conditions:*

- (V1) *every existential $\exists R.D \in \lambda_Q(\sigma)$ has a witness;*
- (V2) *every universal $\forall R.D \in \lambda_Q(\sigma)$ is respected;*
- (V3) *every domain in the quotient has non-empty Need_R -filtered domain;*
- (V4) *the extracted descriptors are support-closed and locally coherent;*
- (V5) *$\equiv_{\text{EQ}}$ is a typed-EQ congruence;*
- (V6) *$C_0 \in \lambda_Q(\sigma_*)$.*

Proof. **(V1) Existential satisfaction.** For $R \in \{\text{DR}, \text{PO}\}$: $\text{Wit}_d(\sigma)$ provides (σ', R) with $D \in \lambda_Q(\sigma')$, by Construction 8.1. Relation-safety holds by Lemma 8.2.

For $R \in \{\text{PP}, \text{PPI}\}$: by Lemma 11.2, some σ' with $\sigma \rightsquigarrow_R \sigma'$ has $D \in \lambda_Q(\sigma')$.

(V2) Universal satisfaction. For $R \in \{\text{DR}, \text{PO}\}$: any sibling-mosaic edge from σ to σ' labelled R is realized by a model pair, which satisfies the universal by Lemma 2.2.

For $R \in \{\text{PP}, \text{PPI}\}$: Lemma 11.4 establishes universal propagation along parent reachability.

(V3) Non-empty filtered domains. For any pair-domain $\text{Dom}(\sigma_1, \sigma_2)$ extracted from the model:

$$\text{Dom}(\sigma_1, \sigma_2) = \{\rho(\text{orig}(u), \text{orig}(v)) : \text{st}_d(u) = \sigma_1, \text{st}_d(v) = \sigma_2\}.$$

Every $R \in \text{Dom}(\sigma_1, \sigma_2)$ is realized by a model pair, hence is relation-safe by Lemma 2.2. Therefore the Need_R -filtered domain equals the unfiltered domain, and is non-empty whenever $\text{Dom}(\sigma_1, \sigma_2)$ is non-empty.

(V4) Coherent descriptors. By Theorem 10.1.

(V5) Typed-EQ congruence. By Lemma 9.3.

(V6) Evaluation state. u_* has $\text{orig}(u_*) = a_0$ and $\lambda(u_*) = \text{tp}_{\mathcal{I}_w}(a_0) \ni C_0$. Hence $C_0 \in \lambda_Q(\sigma_*)$. $\square$

13 Main theorem and decidability

Proof of Theorem 1.1. Given $\mathcal{I}, d_* \models C_0$:

1. Choose a selected witness policy w (Definition 3.1).

2. Pass to the witness-generated submodel $\mathcal{I}_w$ (Lemma 3.3); $a_0 := d_* \in C_0^{\mathcal{I}_w}$.
3. Construct the generated split tree T (Construction 5.1, Lemma 5.3).
4. Define rank- d profiles st_d (Definition 6.1).
5. Form the finite quotient Q (Definition 6.3).
6. Extract sibling descriptors as support-closed mosaics (Construction 7.8, Lemma 7.9).
7. Extract witness menus (Construction 8.1, Lemma 8.2).
8. Verify validity (Theorem 12.1).

The resulting Q is a valid finite generated split-forest quotient for C_0 with $C_0 \in \lambda_Q(\sigma_*)$. $\square$

Proof of Corollary 1.2. $|\Sigma_d|$ is finite by Lemma 6.2. Validity of a candidate quotient Q is decidable: each (V1)–(V6) condition is a finite check on finite data; reachability (V1) for PP/PPI is decidable in $O(|S|^2)$ per eventuality; descriptor coherence (V4) is decidable in time polynomial in $|\text{Mos}|$.

If C_0 is satisfiable, Theorem 1.1 produces a valid quotient. If a valid quotient exists, soundness ([4, Theorem 1.20], [5, Theorem 9.1]) produces a model. Thus C_0 is satisfiable iff a valid finite quotient exists, and enumeration terminates. $\square$

14 Discussion

14.1 Why this is not Büchi acceptance

The key observation that lets us avoid automata-theoretic machinery: every transitive eventuality $\exists R.D$ at $\sigma \in S$ for $R \in \{\text{PP}, \text{PPI}\}$ is discharged by ordinary reachability over the finite directed graph (S, par_Q) .

In an unfolding of Q , an infinite PP-branch corresponds to a walk in (S, par_Q) . Once the walk enters a strongly connected component (SCC), it cycles through that component forever (because $|S|$ is finite). Eventually-recurrent profiles on the SCC are exactly the profiles reachable from any profile in the SCC.

So: a PP-eventuality $\exists \text{PP}.D$ at σ is realized in the unfolding iff D appears in some profile reachable from σ . This is decidable by:

1. BFS from σ in (S, par_Q) ;
2. check whether any reachable profile σ' has $D \in \lambda_Q(\sigma')$.

GPT-5.5’s stem-and-cycle phrasing (“ $u_0, \dots, u_m, (v_0, \dots, v_{k-1})^\omega$ ”, [5, Definition 6.4]) is one specific way to present this realization condition. But the underlying check is the same finite reachability question, and our formulation in Lemma 11.2 is closer to standard description logic blocking conditions.

14.2 Comparison with v1

Concern	v1 (April 2026)	v2 (this paper)
Starting model	arbitrary PP-DAG with assumed $\leq_{\text{PP-min}}$	witness-generated submodel
Cover edges	Hasse covers of ambient order	generated cover edges in $D_w(a_0)$
Ambient root	required (added freshly if needed)	not required
Self-loops on parent	not permitted	permitted (handles $C \uparrow$)
Sibling descriptor	raw pair-tables	support-closed mosaics
Triple coherence (C5)	witness conflation across occurrences	joint-realization within a mosaic
Typed-EQ congruence	implicit via orig map	explicit axiom
PP/PPI eventualities	“ancestor in tree” (vague)	finite graph reachability

14.3 Comparison with GPT-5.5’s automata version

GPT-5.5’s coherent generated split-forest proof [5] arrives at the same conclusion via the same structural ideas (witness-generated submodels, generated covers, support-closed mosaics, typed-EQ congruence) but formalizes the eventuality discharge as a Büchi-style cycle-realization condition.

Our v2 differs only in presentation: where GPT-5.5 writes $u_0 \cdots u_m(v_0 \cdots v_{k-1})^\omega$ and requires every eventuality on the cycle to be met on the cycle, we write “every $\exists \text{PP}.D$ at σ has a reachable witness profile,” which is finite-graph reachability. These are equivalent: in the finite quotient, “met somewhere on the ω -trace” equals “reachable in par_Q .”

The benefit of the reachability formulation: it stays inside the standard description-logic toolbox (similar to fixpoint blocking conditions in μ -calculus tableaux), and avoids invoking ω -language theory.

14.4 Honest assessment

We flag the following points for scrutiny:

Witness policy is non-canonical. Different selected witness policies produce different generated submodels and different split trees. The extraction works for *any* fixed policy. We do not claim a canonical choice.

Profile bound is not elementary. $|\Sigma_d|$ is bounded by a tower of exponentials in $|\text{cl}(C_0)|$. Decidability follows, but tight complexity remains open. The PO-coherent fragment achieves a 2-EXPTIME quotient bound [8].

Mosaic vertex bound. Construction 7.8 restricts to mosaics on $\leq |\text{cl}(C_0)| + 4$ vertices. This is sufficient for support-closure (SC1)–(SC4), but a fully rigorous treatment would derive the bound from a Ramsey-type argument on closure size. GPT-5.5 sketches such an argument in [5, Lemma 8.4].

Reachability vs strongly-connected components. Lemma 11.2 establishes existence of a reachable witness profile for each eventuality. The full correctness argument requires that *simultaneous* discharge of all eventualities on a branch is realized by a single coherent infinite walk. This follows from finiteness of S and the RCC5 patchwork property: the SCC of σ in (S, par_Q) contains all profiles eventually visited from σ , and any ordering of the eventualities can be interleaved on the SCC walk.

None of these points appears to harbour a genuine gap. The extraction rests on three clean principles:

1. witness-generated submodels are sparse (Lemma 3.3);
2. model edges are relation-safe (Lemma 2.2);
3. finite-graph reachability discharges transitive eventualities (Lemma 11.2).

15 Conclusion

We have written out the completeness extraction for $\mathcal{ALCI}_{\text{RCC5}}$ in a repaired form that fixes the two structural defects of the v1 draft: the unsound Hasse construction and the witness-conflated triple-coherence proof. The repair uses witness-generated submodels, generated cover edges, support-closed mosaics, an explicit typed-EQ congruence, and finite-graph reachability discharge.

Crucially, the repair stays inside the original split-forest framework: vertical PP/PPI branches, sibling-interface descriptors, and finite quotient enumeration are all preserved. The cycle-realization condition that motivates GPT-5.5’s automata-flavoured proof reduces here to ordinary reachability over the finite quotient graph, sidestepping any need for Büchi acceptance or ω -language theory.

Combined with the soundness chain of [4, 5], this yields:

Concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ under strong EQ semantics is decidable.

The exact complexity is open. The quotient search is decidable but non-elementary; an EXPTIME bound (matching the known $\mathcal{ALCI}$ lower bound) would require a tighter analysis of the rank- d profile space.

Computational evidence. An independent implementation [9] cross-validates the cover-tree tableau (the operational counterpart of the split-forest quotient search) against a quasimodel-based reasoner on 911 test concepts with zero mismatches. A decomposition test confirms that 775/775 satisfiable concepts (100%) have finite cover-tree models across 12.2 M enumerated models, with zero genuine counterexamples [10, 11].

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